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electronics
FOR YOU

Q1. I want to make a sensor-based vacuum cleaner that changes its direction when it detects any obstacle. I searched all EFY magazines for such a project, but in vain. Please help me!

Amardeep Singh
Karnal

A1. The automatic vacuum cleaner robot (see Fig. 1) is an interest-



Fig. 1: Automatic vacuum cleaner (Courtesy: Wikipedia)

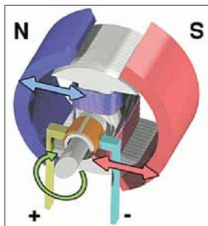


Fig. 2: DC motor (Courtesy: Wikipedia)

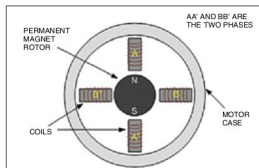


Fig. 3: Two-phase permanent magnet type stepper motor (Courtesy: Wikipedia)

ing project that can be implemented fairly easily. The complete design can be divided into five major sections: Mechanics, motors and driver section, controlling unit, sensing unit and software. The design complexity depends on the level of intelligence you would like to put into your robot. This complexity will mainly relate to the sensing unit and software.

Mechanics. The overall mechanical design can be circular with wheels attached to the circular axis. This way the robot can spin in place in any direction. Such a design needs to be stabilised with another freely moving wheel at the back (it should not be driven by any motor). If you are making this project only for proof-of-concept, you can use a simple cleaning assembly with rotating brush underneath the unit to clean the floor. This brush can be driven by a DC motor (see

Fig. 2).

Motors and driver section. Here you need to decide which motor to use for moving the robot and rotating the brush. I would recommend stepper motors (see Fig. 3) to move the robot.

DC motors are not precise and it would be difficult to find out how far the wheels have travelled. Also, additional circuitry like H-Bridge would be required to drive DC motors back and forth.

Stepper motors, on the other hand, are precisely controlled and can move backward and forward just by using ULN2003. With stepper motors, it is theoretically possible to calculate the distance travelled by the wheel as also the room size. These values can be entered in the algorithm to better cover the room.

You might encounter lack of torque in stepper motors. A workaround would be the use of higher-capacity batteries.

For rotating the brush, the DC motor should be interfaced